Uncertainty Prediction Plots (1FA006)

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The Jupyter notebook can be found in this GitHub repository.

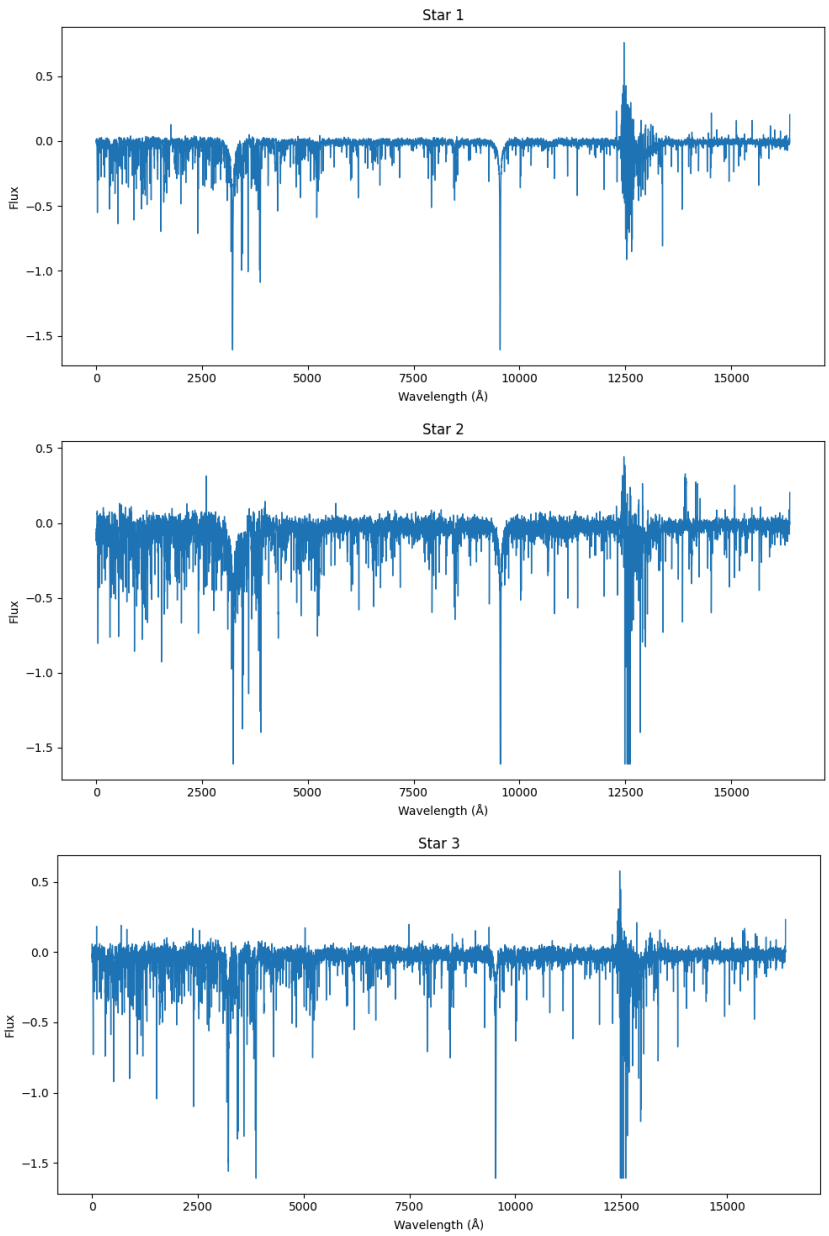
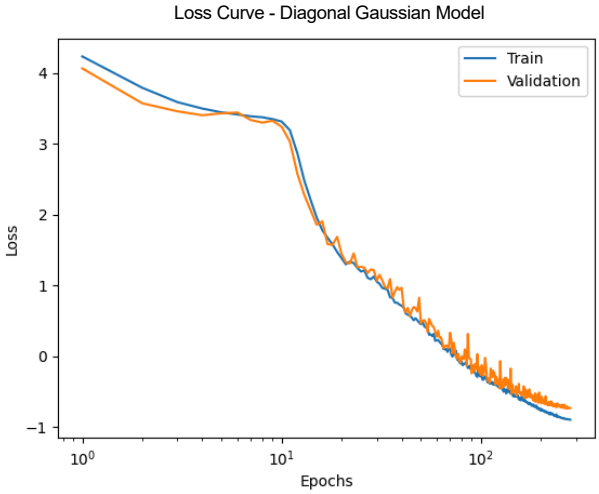
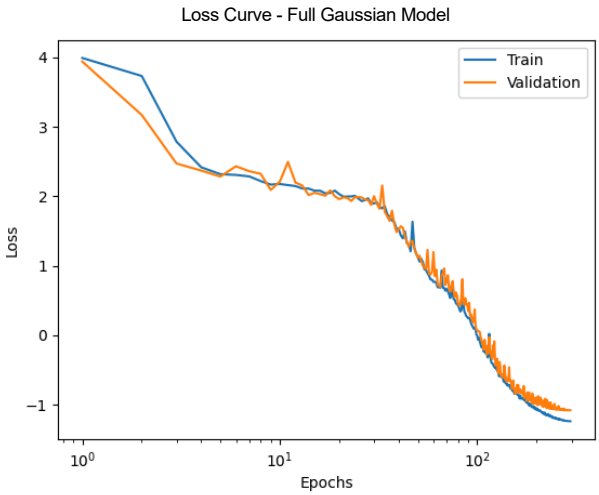


Figure 1: Examples of stellar spectra from the GALAH DR4 Dataset.





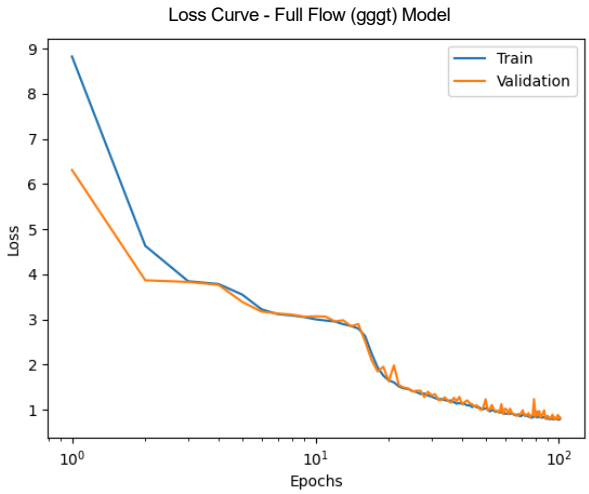


Figure 2: Loss Curves for all three models. All models show stable training without major overfitting.

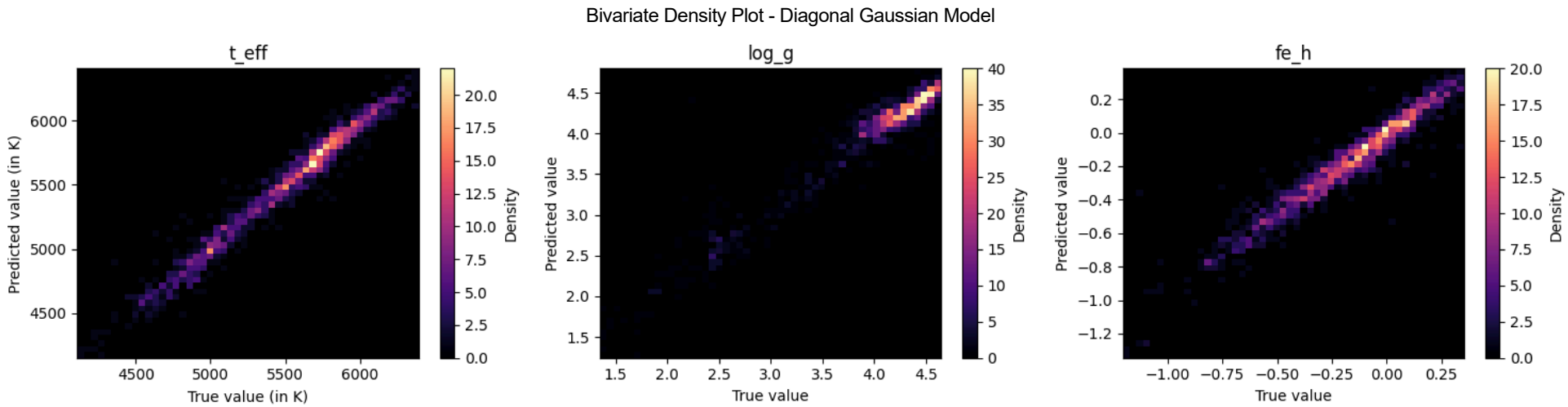


Figure 3: True v/s Predicted means comparision for the Diagonal Gaussian model. The predictions follow the y=x trend closely, showing good accuracy.

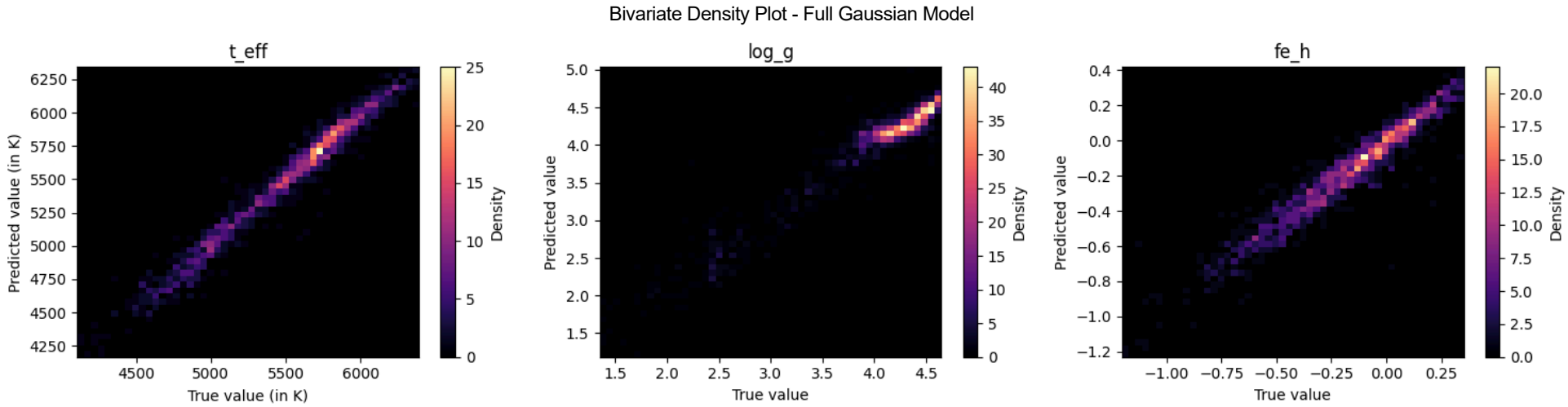


Figure 4: True v/s Predicted means comparision for the Full Gaussian model. There is slightly more scatter, especially at lower temperatures and lower metallicity. Overall, the accuracy still seems to be good.

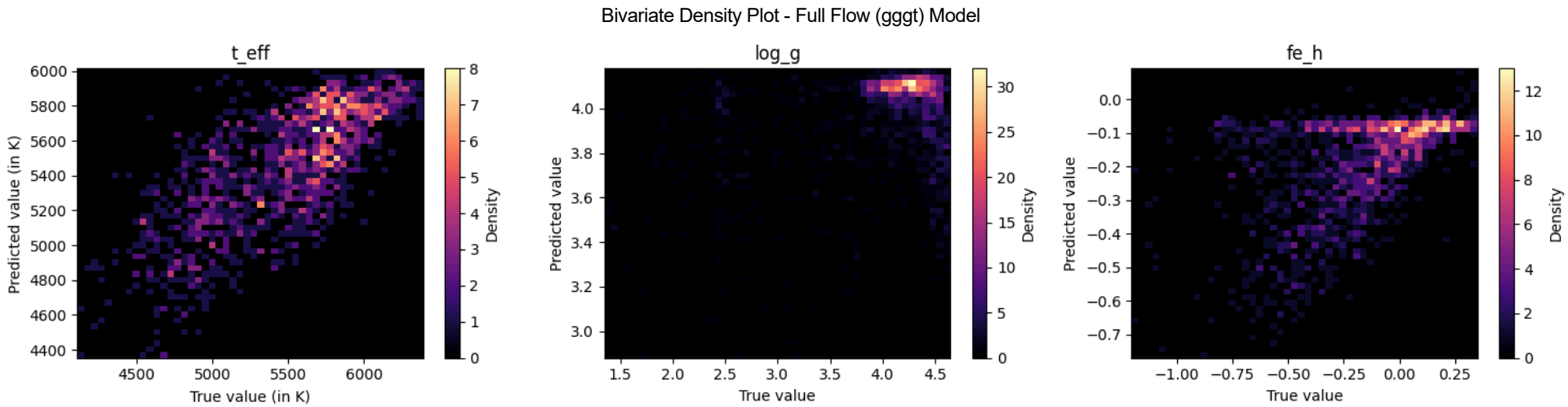


Figure 5: True v/s Predicted means comparision for the Full Flow (gggt) model. The density is scattered, showing poor accuracy.

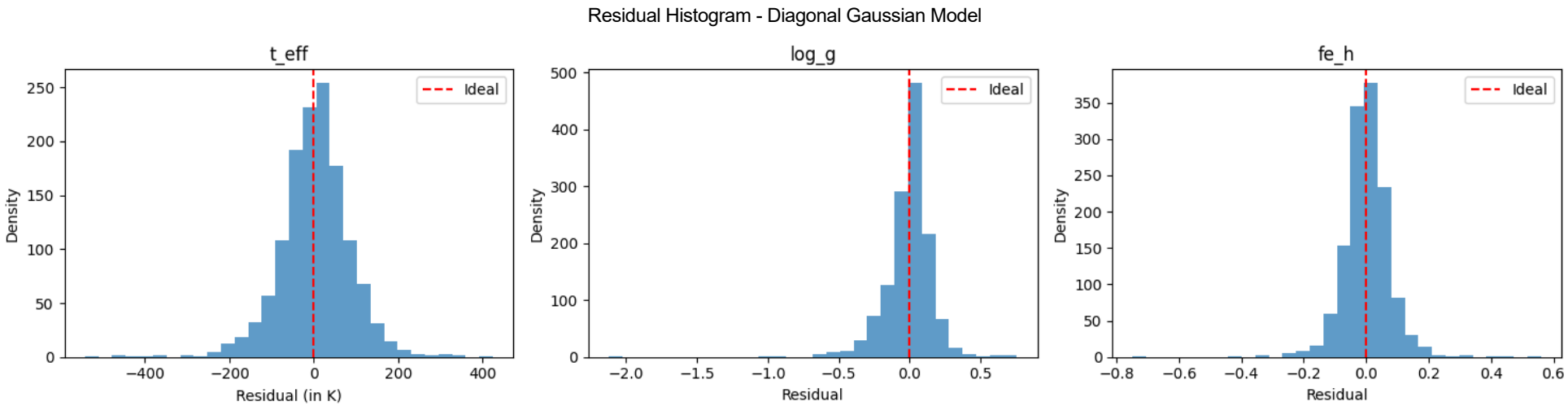


Figure 6: Residual Histograms for Diagonal Gaussian Model. The residuals are centered around zero and the distribution is narrow. This indicates low bias.

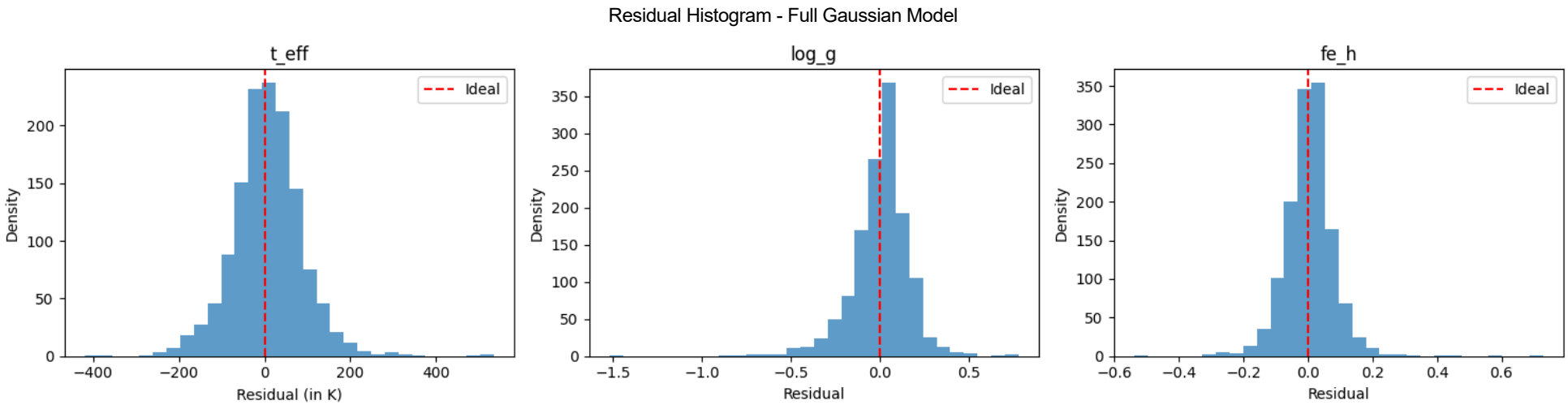


Figure 7: Residual Histograms for Full Gaussian Model. The residuals are slightly wider compared to the Diagonal Gaussian model.

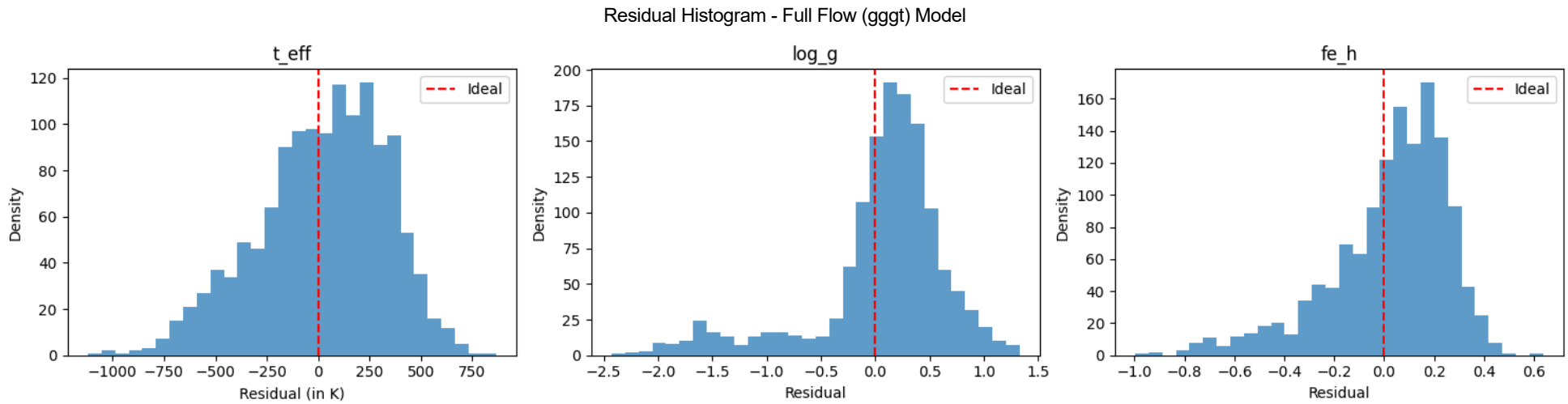


Figure 8: Residual Histograms for Full Flow (gggt) Model. The residuals have longer tails and the distribution is wide. This means that there are errors in prediction.

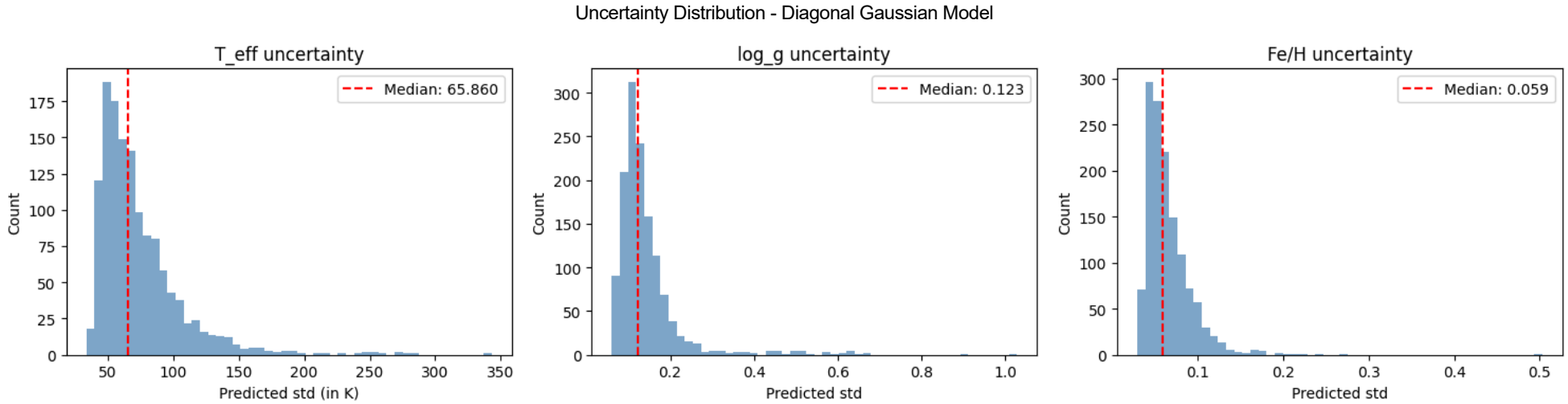


Figure 9: Distributions of predicted mean of standard deviations (uncertainties) for the Diagonal Gaussian model. The distribution is somewhat sharp peaked and also approximately matches the residual distribution as well. The median tells the ”typical” uncertainty that the model assigns to the predicted mean. Median is used instead of mean to account for the asymmetric distribution.

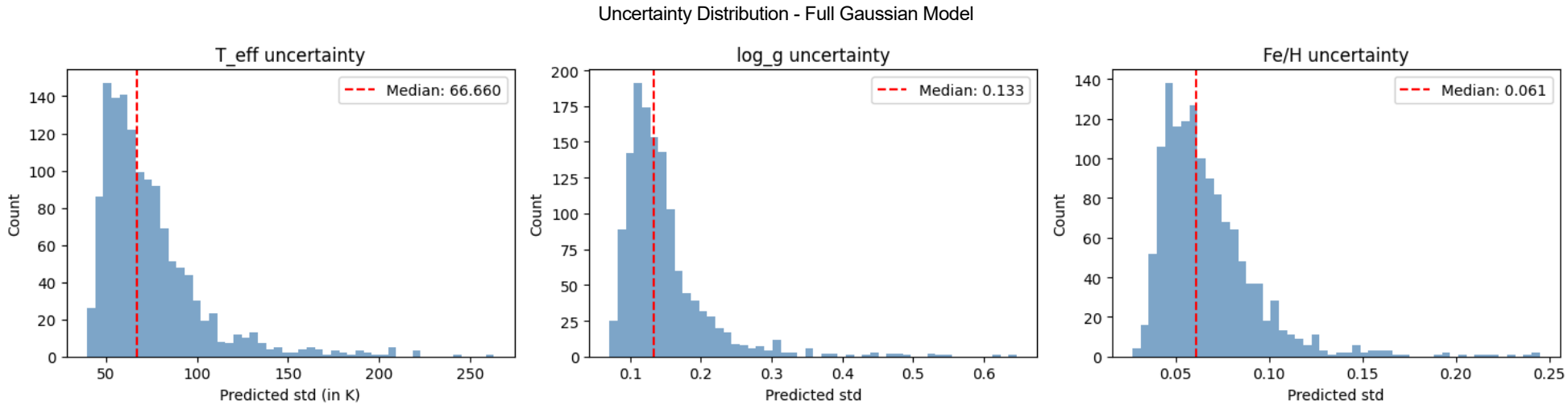


Figure 10: Distributions of predicted mean of standard deviations (uncertainties) for the Full Gaussian model. The ”typical” uncertainties are slightly higher than Diagonal Gaussian model, as seen in the broader distribution.

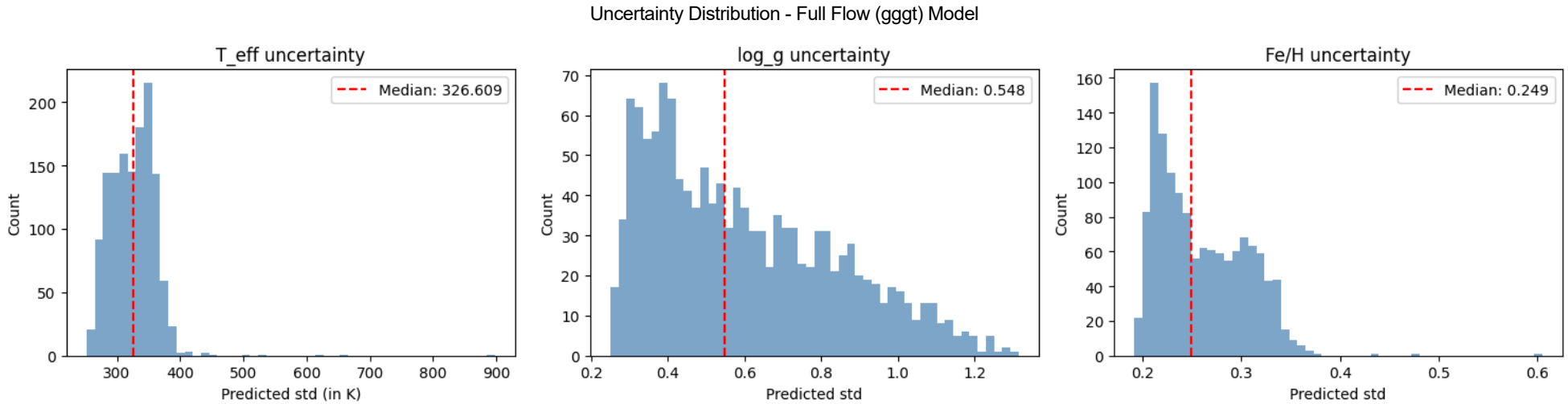


Figure 11: Distributions of predicted mean of standard deviations (uncertainties) for the Full Flow (gggt). The distribution is extremely wide and the ”typical” uncertainties are too large. This shows that the model fails to predict the labels with certainty.

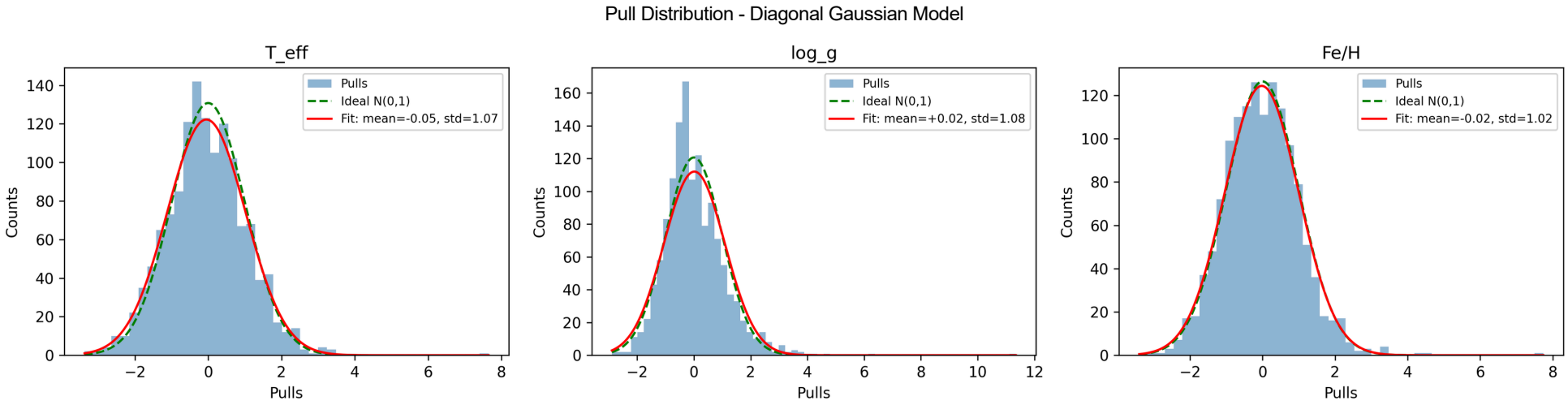


Figure 12: Pull Distribution for Diagonal Gaussian model. The pulls roughly match the ideal shape and the mean and standard deviation values are close to the ideal case.

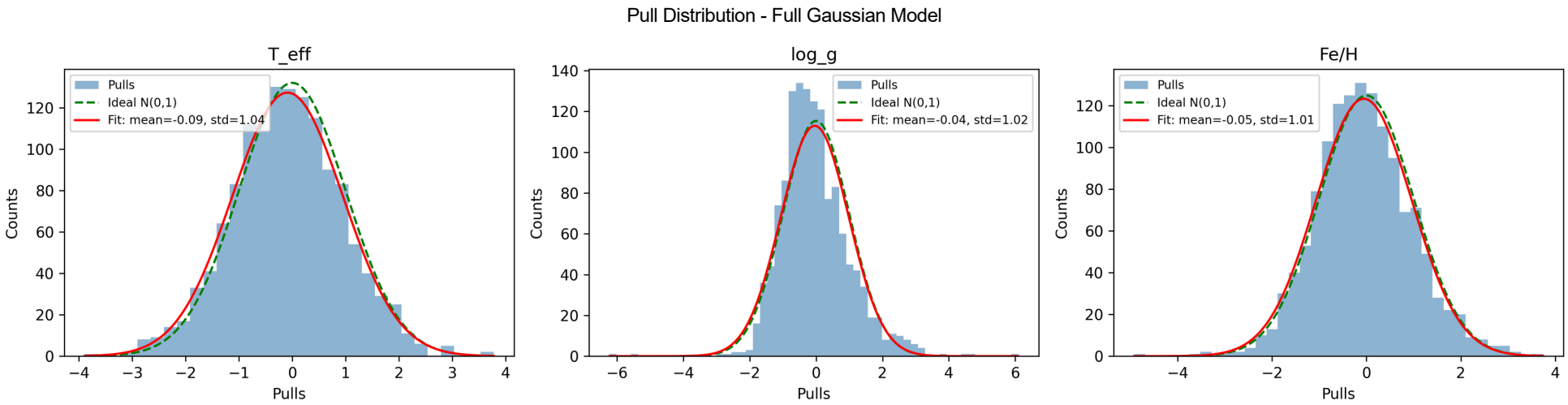


Figure 13: Pull Distribution for Full Gaussian model. The pulls roughly match the ideal shape and the mean and standard deviation values are close to the ideal case.

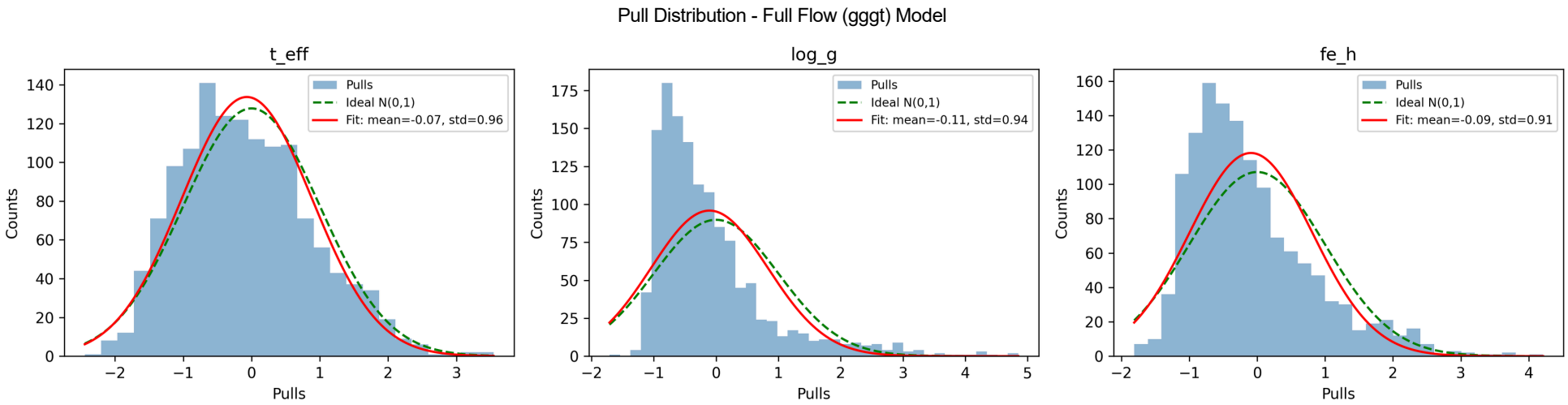


Figure 14: Pull Distribution for Full Flow (gggt) model. There is a visible mismatch between the ideal normal distribution and the fitted one.

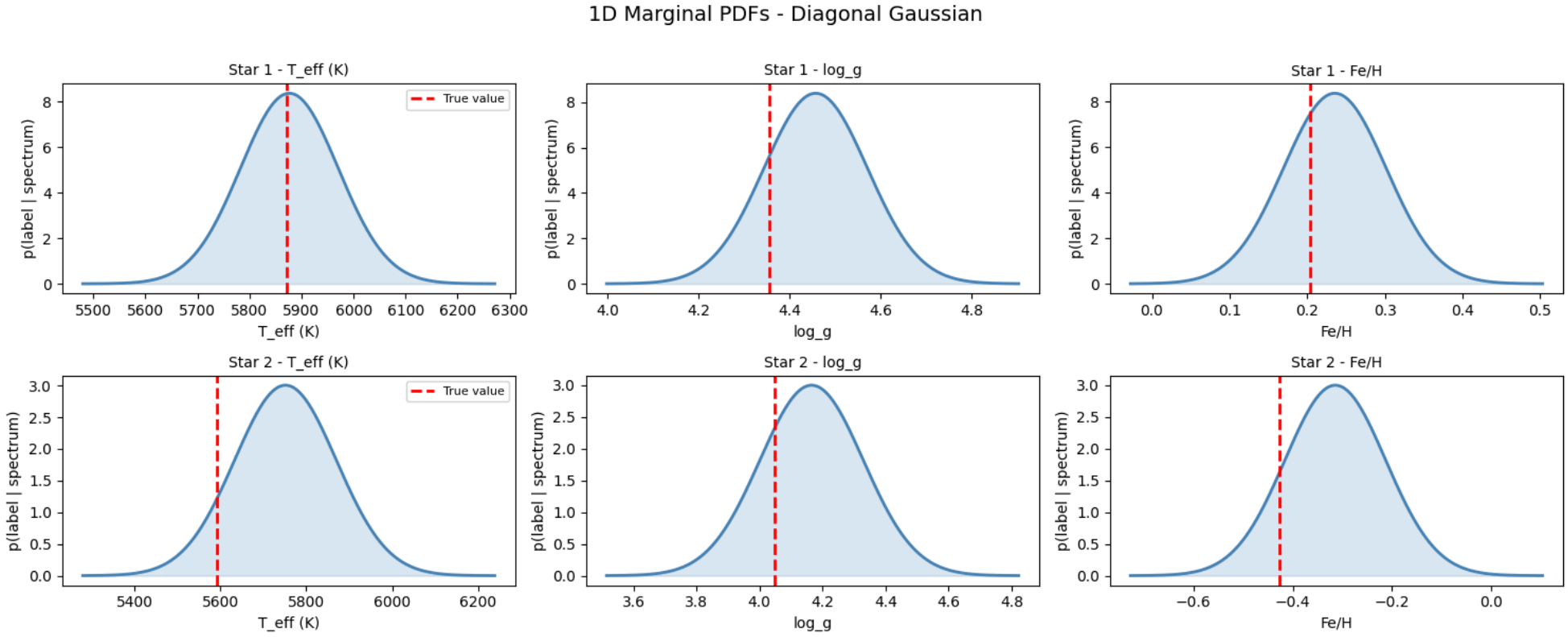


Figure 15: 1D marginal PDF for two example stars using the Diagonal Gaussian model.

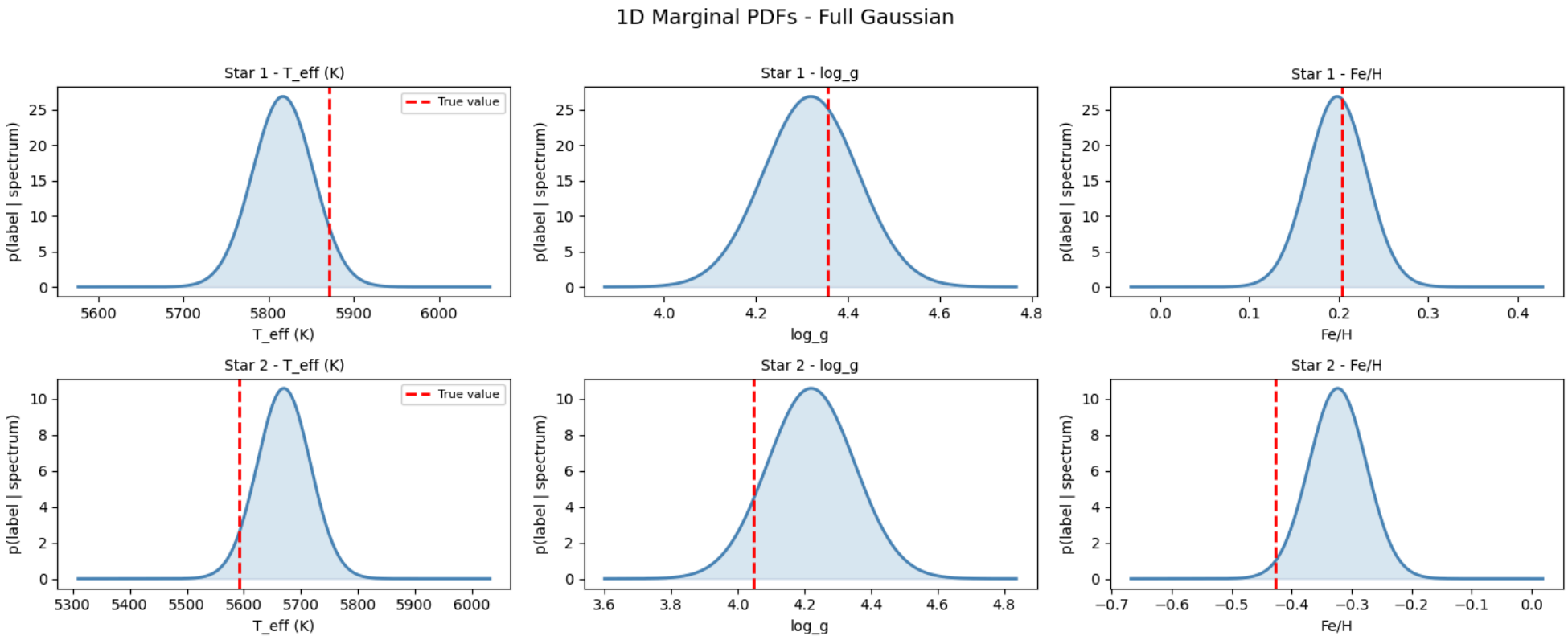


Figure 16: 1D marginal PDF for two example stars using the Full Gaussian model.

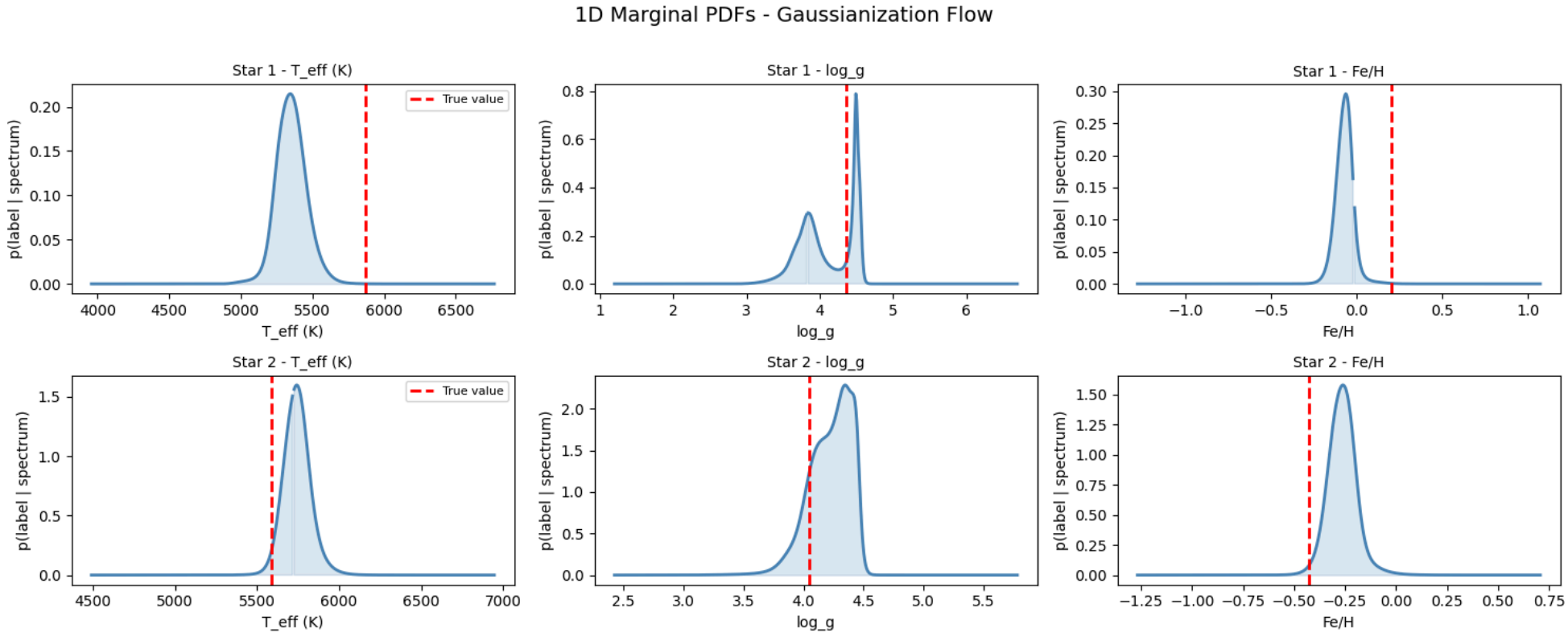


Figure 17: 1D marginal PDF for two example stars using the Full Flow (gggt) model. We can see that the trained PDFs are bimodal in some cases.